

Alzheimer's Disease and Aging: Aims 1 and 2

Project 2

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Aim 1 and Aim 2 Overview

- Aim 1: Are inflammatory markers (IL-6, TNF- α , MCP-1, Eotaxin-1) associated with 1 year changes in episodic memory and cortical thickness?
 - Hypothesis 1a: Baseline inflammatory markers
 - Hypothesis 1b: 1 year change inflammatory markers
- Aim 2: Does amyloid deposition modify these associations?
 - Primary Hypothesis: Baseline inflammatory markers
 - Secondary Hypothesis: 1 year change inflammatory markers

Data Analysis Plan

Aim 1: $\beta_1 \neq 0$

- $$\text{Outcome}_{Y1-Y0} = \beta_0 + \beta_1 \text{Marker}_{Y1 \text{ or } Y1 - Y0} + \beta_2 \text{Outcome}_{Y0} + \beta_n \text{Covariates} + \varepsilon$$

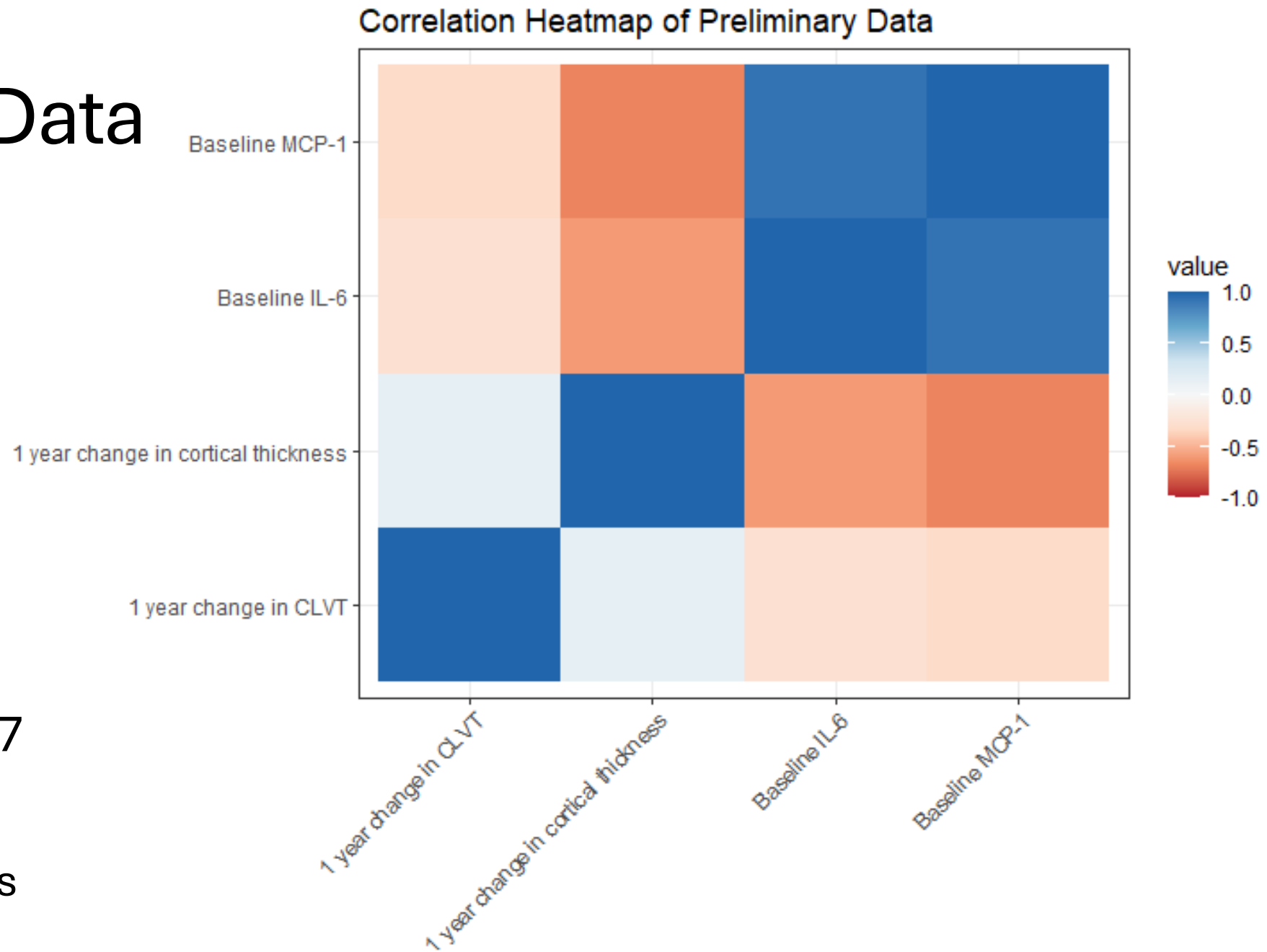
- Covariates will be evaluated as precision variables in an exploratory manner
 - Age, sex, APOE status added *a priori*
- Missing data will be investigated for bias
- Type I Error will be controlled as 0.05/4 (markers), 0.05/6 (outcomes), or 0.05/24
 - More formal adjustments will be explored once empirical correlations are known

Aim 2: $\beta_3 \neq 0$

- $$\text{Outcome}_{Y1-Y0} = \beta_0 + \beta_1 \text{Marker}_{Y1 \text{ or } Y1 - Y0} + \beta_2 \text{Amyloid}_{Y0} + \beta_3 \text{Marker}_{Y1 \text{ or } Y1 - Y0} * \text{Amyloid}_{Y0} + \beta_4 \text{Outcome}_{Y0} + \beta_n \text{Covariates} + \varepsilon$$

Preliminary Data

- Preliminary data included two change outcomes and two baseline inflammatory markers
- Correlations between markers and outcomes ranged from 0.2-0.7
 - Informed effect size range for power calculations



Aim 1 Power

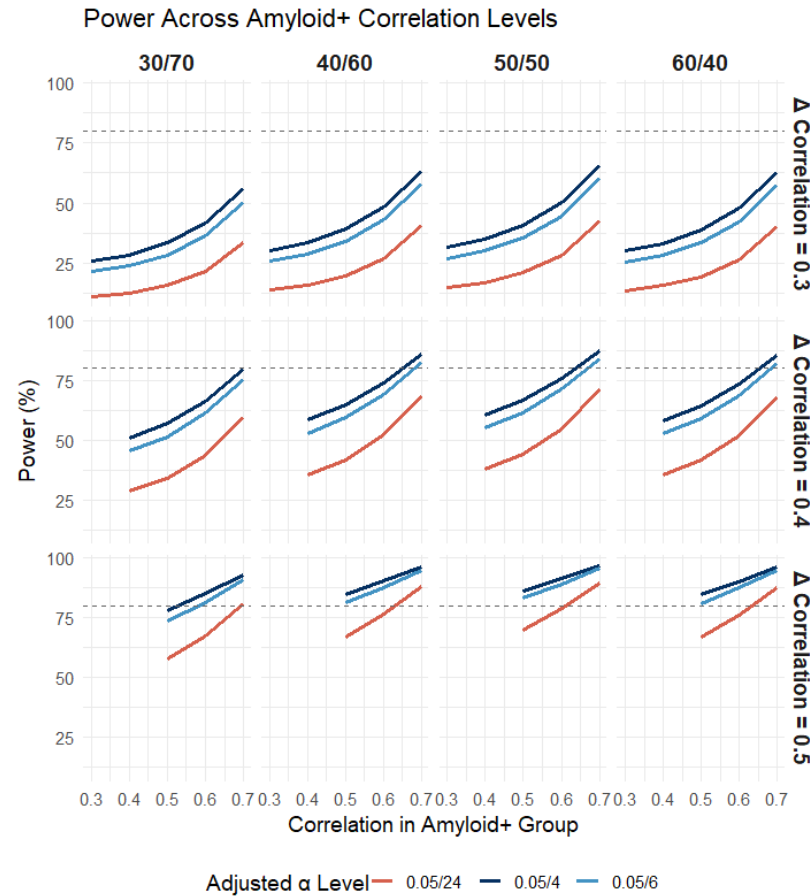
- Power > 80% for correlations ≥ 0.30 across all α levels
- Aim 1 is well-powered for moderate effects

Power to Detect Correlations Across α Levels

Correlation	Power ($\alpha = 0.05/4$)	Power ($\alpha = 0.05/6$)	Power ($\alpha = 0.05/24$)
0.20	56.7%	51.1%	34.0%
0.25	80.6%	76.5%	61.1%
0.30	94.2%	92.4%	84.0%
0.35	99.0%	98.5%	95.8%
0.40	99.9%	99.8%	99.4%
0.45	100.0%	100.0%	100.0%
0.50	100.0%	100.0%	100.0%

Aim 2 Power

- Large interactions (0.50) detectable with moderate amyloid+ correlations
- Moderate interactions (0.40) require strong correlations
- Under $\alpha = 0.05/24$, only very strong effects are detectable



Minimum Amyloid+ Correlation Needed for 80% Power

Required Correlation in Amyloid+ Group	Group Difference (Δ correlation)	Adjusted α Level	Power (%)
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30/70 Split (Low Prevalence)

0.70	0.50	$\alpha = 0.05/24$	81.00
0.60	0.50	$\alpha = 0.05/4$	84.90
0.60	0.50	$\alpha = 0.05/6$	81.30

40/60 Split (Moderate Imbalance)

0.70	0.40	$\alpha = 0.05/4$	85.80
0.70	0.40	$\alpha = 0.05/6$	82.40
0.70	0.50	$\alpha = 0.05/24$	88.00
0.50	0.50	$\alpha = 0.05/4$	84.60
0.50	0.50	$\alpha = 0.05/6$	81.00

50/50 Split (Balanced Groups)

0.70	0.40	$\alpha = 0.05/4$	87.30
0.70	0.40	$\alpha = 0.05/6$	84.20
0.70	0.50	$\alpha = 0.05/24$	89.60
0.50	0.50	$\alpha = 0.05/4$	86.20
0.50	0.50	$\alpha = 0.05/6$	82.90

60/40 Split (Moderate Imbalance)

0.70	0.40	$\alpha = 0.05/4$	85.50
0.70	0.40	$\alpha = 0.05/6$	82.00
0.70	0.50	$\alpha = 0.05/24$	87.70
0.50	0.50	$\alpha = 0.05/4$	84.40
0.50	0.50	$\alpha = 0.05/6$	80.80

Discussion

- Interpretation
 - Aim 1: Powered for moderate associations between inflammatory markers and cognitive decline
 - Aim 2: Likely detectable only for large interaction effects
- Limitations
 - Known correlations in predictors, covariates, and outcomes
 - Limited preliminary data
 - Few known empirical correlations
 - No preliminary data for amyloid deposition
 - α thresholds may be too conservative
 - Formal adjustments will be explored when empirical correlations are known.